



Lab 1

Basic Logic Gate Implementation

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Objectives

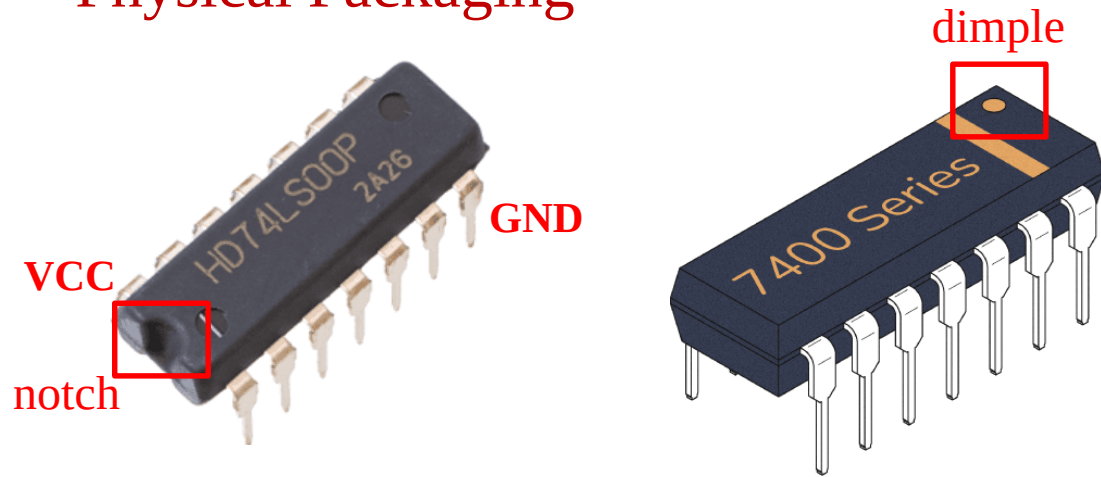
- Understand following components
 - Transistor-Transistor Logic (TTL) ICs 7400, 7404, 7408, 7432
 - LEDs, toggle switches
- Implement **logic circuits** to verify the corresponding **truth tables**

TTL 7400 (NAND)

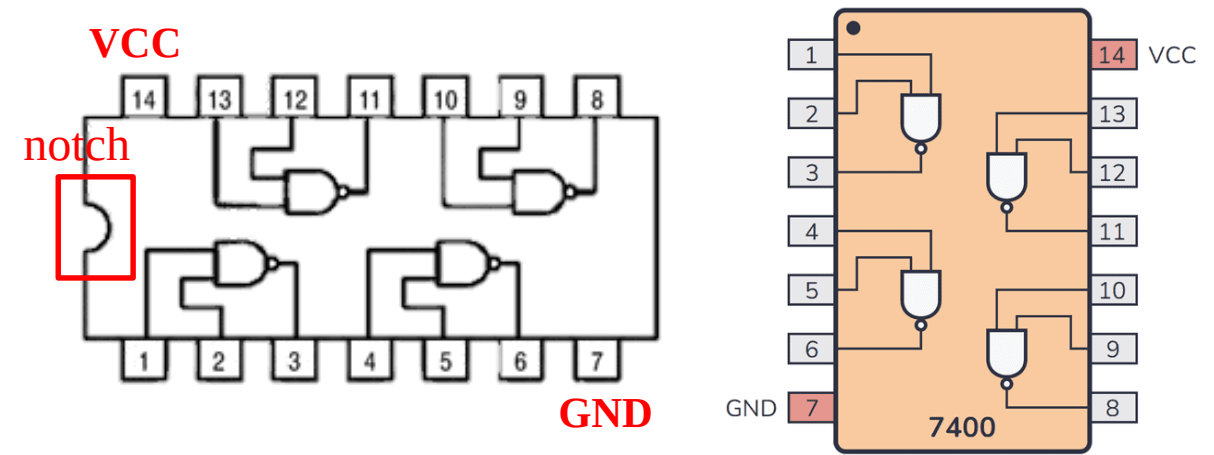
Figure Sources:

- <https://www.igelectronics.com/products/g69b66b32e/1726810000000161177>
- <https://www.build-electronic-circuits.com/7400-series-integrated-circuits/74hc00-74ls00/>

Physical Packaging



Internal Structure



Graphic Symbol



Truth Table

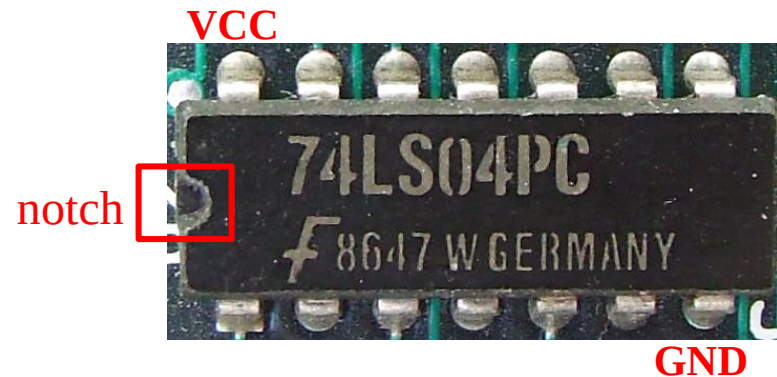
Input A	Input B	Output OUT
0	0	1
0	1	1
1	0	1
1	1	0

TTL 7404 (NOT)

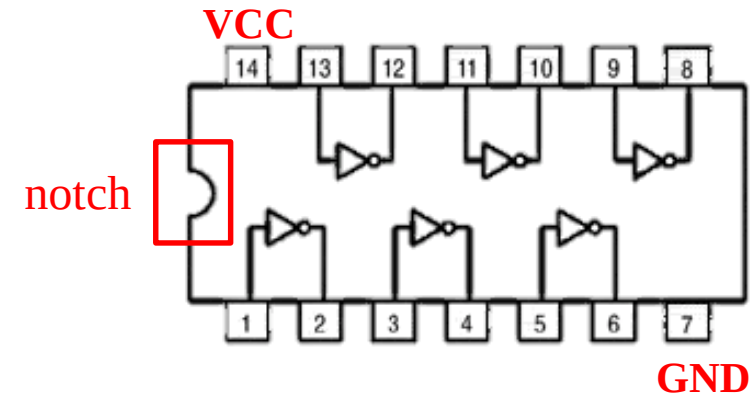
Figure Source:

1. https://pt.wikipedia.org/wiki/TTL_7404

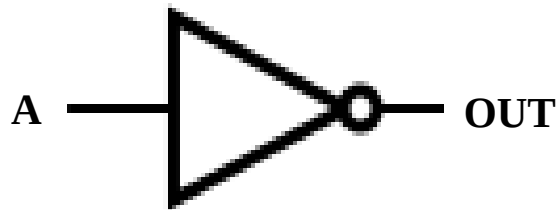
Physical Packaging



Internal Structure



Graphic Symbol



Truth Table

Input A	Output OUT
0	1
1	0

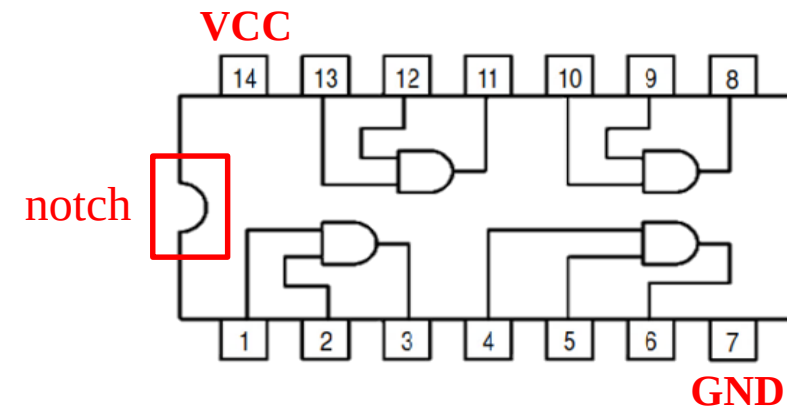
TTL 7408 (AND)

Figure Source:
1. https://pt.wikipedia.org/wiki/TTL_7408

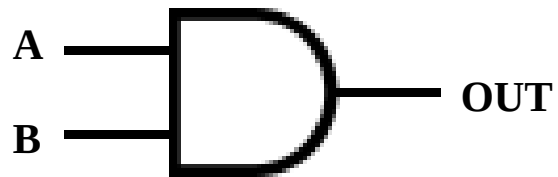
Physical Packaging



Internal Structure



Graphic Symbol



Truth Table

Input A	Input B	Output OUT
0	0	0
0	1	0
1	0	0
1	1	1

TTL 7432 (OR)

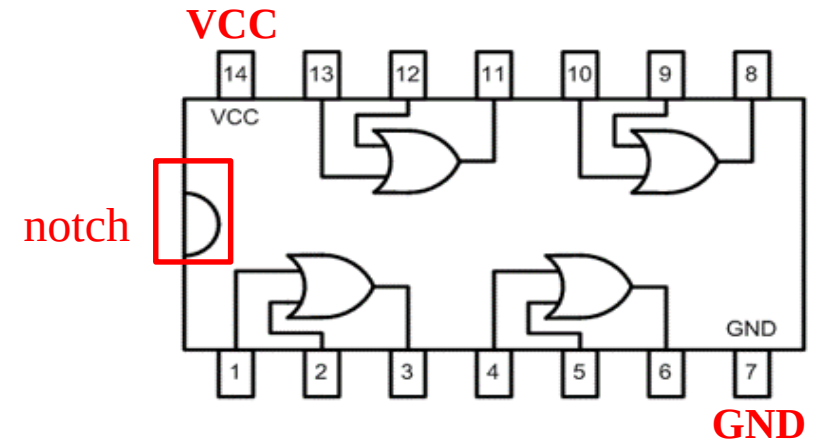
Figure Source:

<https://docs.cirkitdesigner.com/component/e39cedd2-1d38-4573-829f-9f133c89a3b8/7432-or-gate-ic>

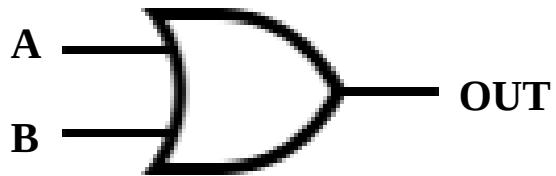
Physical Packaging



Internal Structure



Graphic Symbol



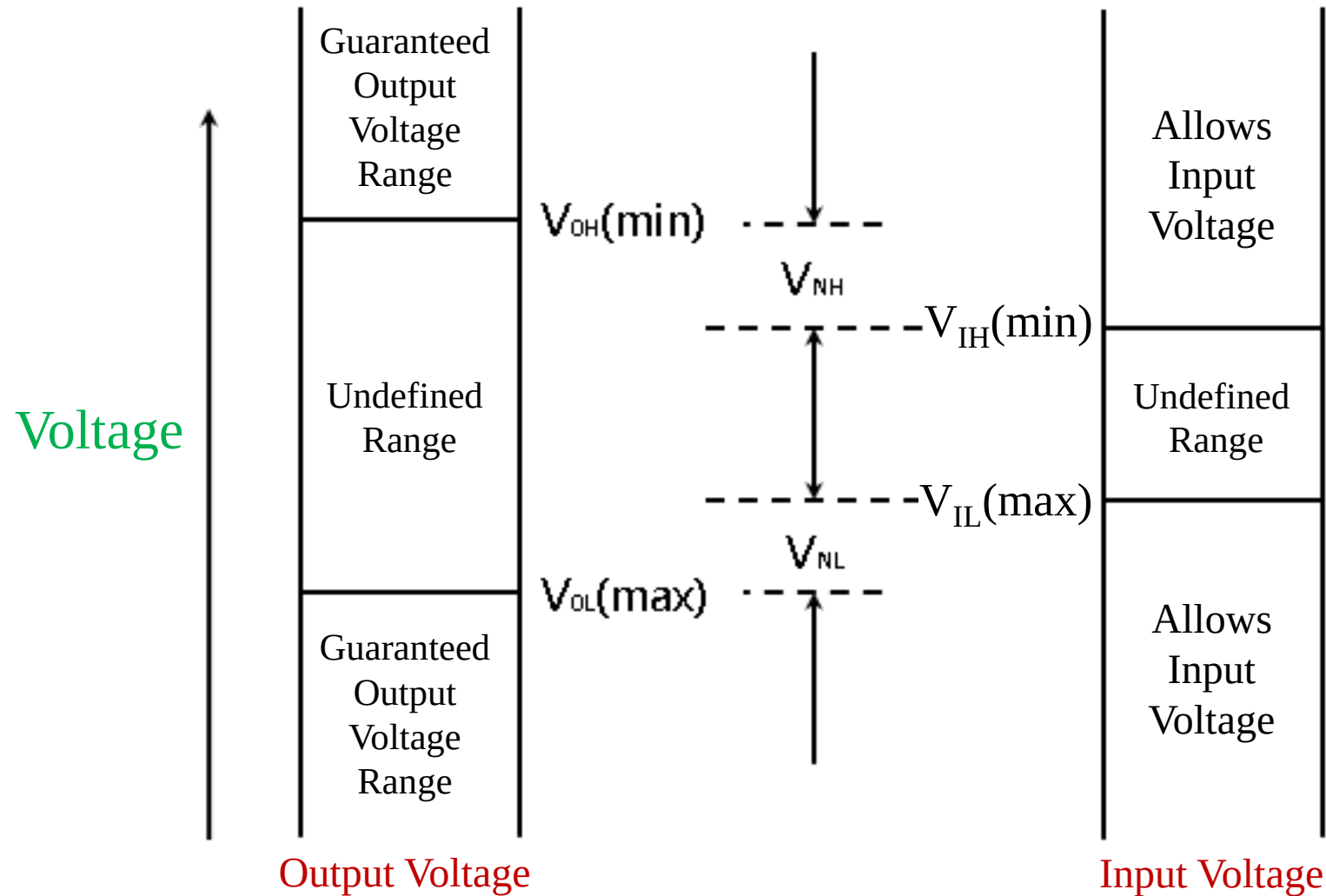
Truth Table

Input A	Input B	Output OUT
0	0	0
0	1	1
1	0	1
1	1	1

Number/Alphabet vs Logic State

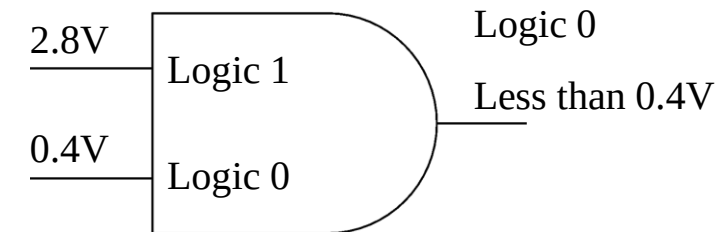
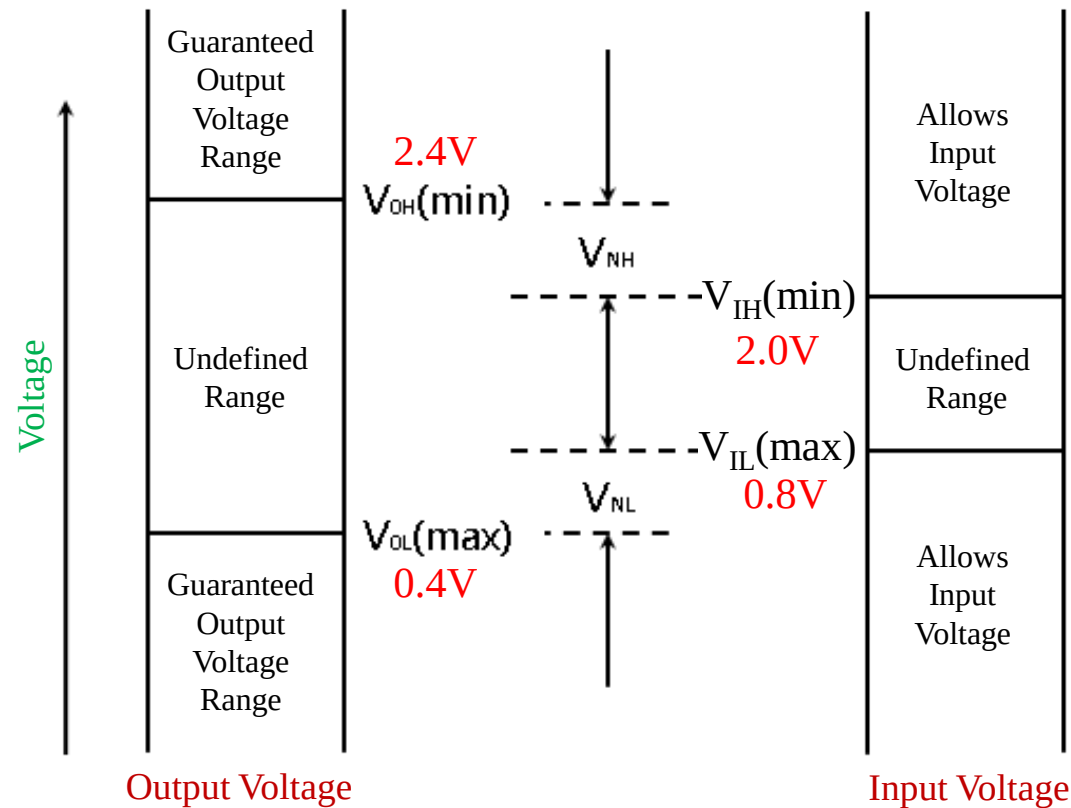
- 0 → Logic LOW (Ground)
- 1 → Logic HIGH (5V)
- x or X → Undefined
- z or Z → High Impedence or Floating
- When an input pin of a TTL IC is floating, it is typically interpreted as a logic HIGH (1).

Voltage Logic Levels



Example of Voltage Logic Levels --- Standard TTL

TTL Type	IC Number	$V_{OH}(\text{min})$	$V_{IH}(\text{min})$	$V_{OL}(\text{max})$	$V_{IL}(\text{max})$	Propagation Delay	Power Consumption
Standard	74xx	2.4V	2.0V	0.4V	0.8V	9 ns	10 mW



Comparison Table

TTL Type	IC Number	$V_{OH}(\text{min})$	$V_{IH}(\text{min})$	$V_{OL}(\text{max})$	$V_{IL}(\text{max})$	Propagation Delay	Power Consumption
Standard	74xx	2.4V	2.0V	0.4V	0.8V	9 ns	10 mW
Low Power	74Lxx	2.4V	2.0V	0.3V	0.8V	33 ns	1 mW
High-Speed	74Hxx	2.4V	2.0V	0.4V	0.8V	6 ns	23 mW
Schottky (high-speed with the same power consumption)	74Sxx	2.4V	2.0V	0.5V	0.8V	3 ns	23 mW
Low Power Schottky	74LSxx	2.4V	2.0V	0.5V	0.8V	9.5 ns	2 mW

Circuit Wiring Comparison

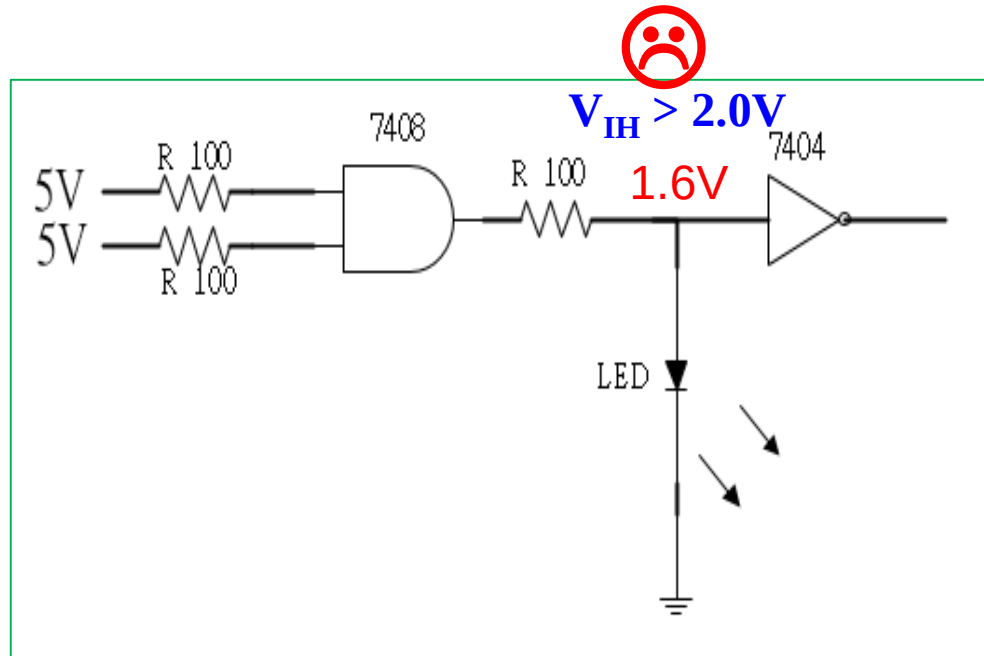


Figure 1

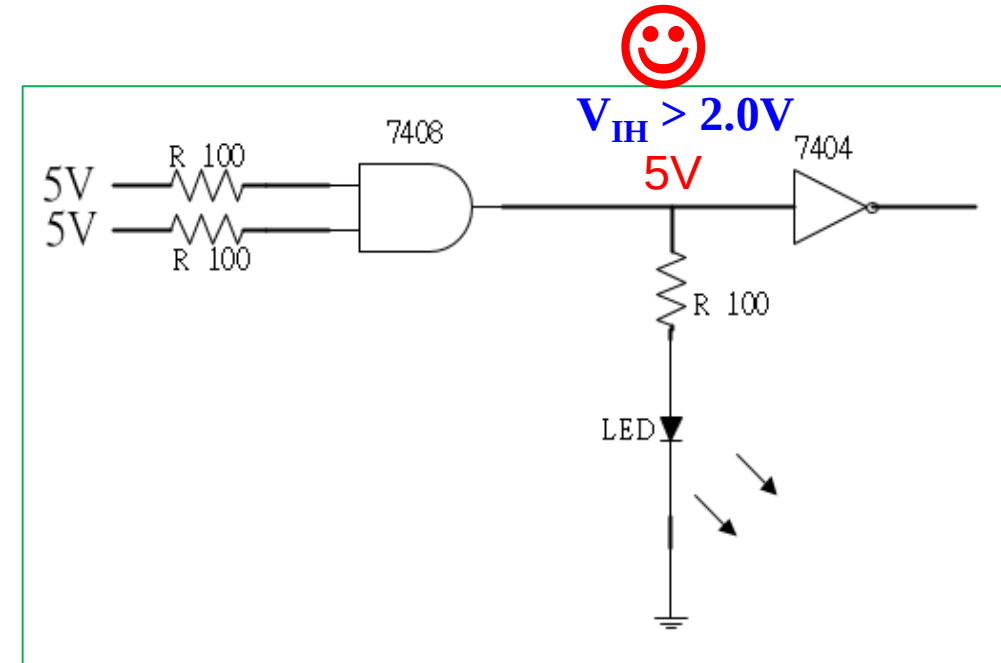


Figure 2

Toggle Switch (DIP Switch) (指撥開關)

- A manually adjustable switch, usually grouped in sets of several switches.
- Typically appears in the standard Dual In-line Package (DIP) (雙列直插封裝) form.

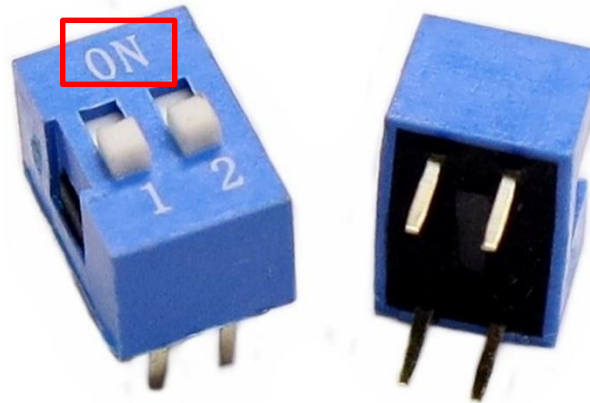
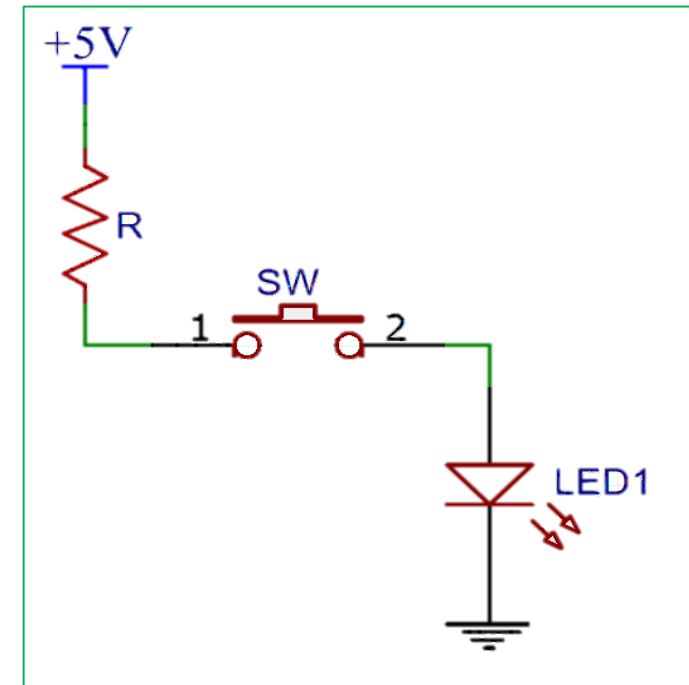
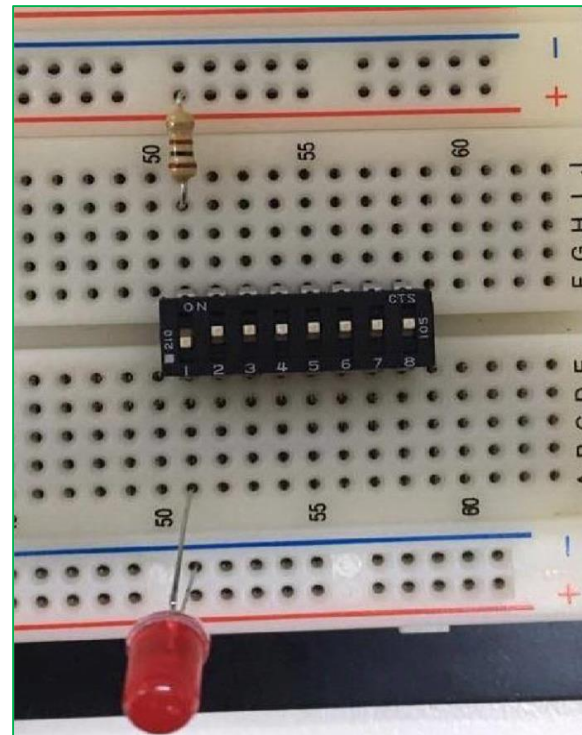
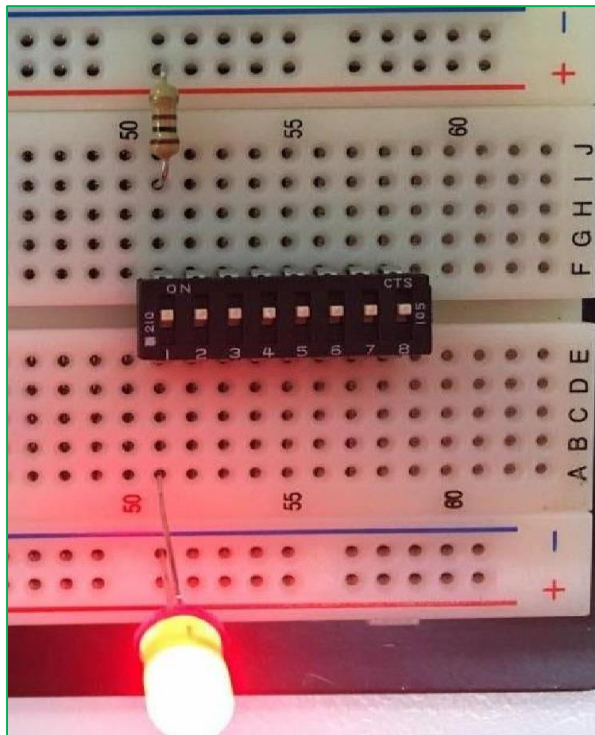


Figure Source:

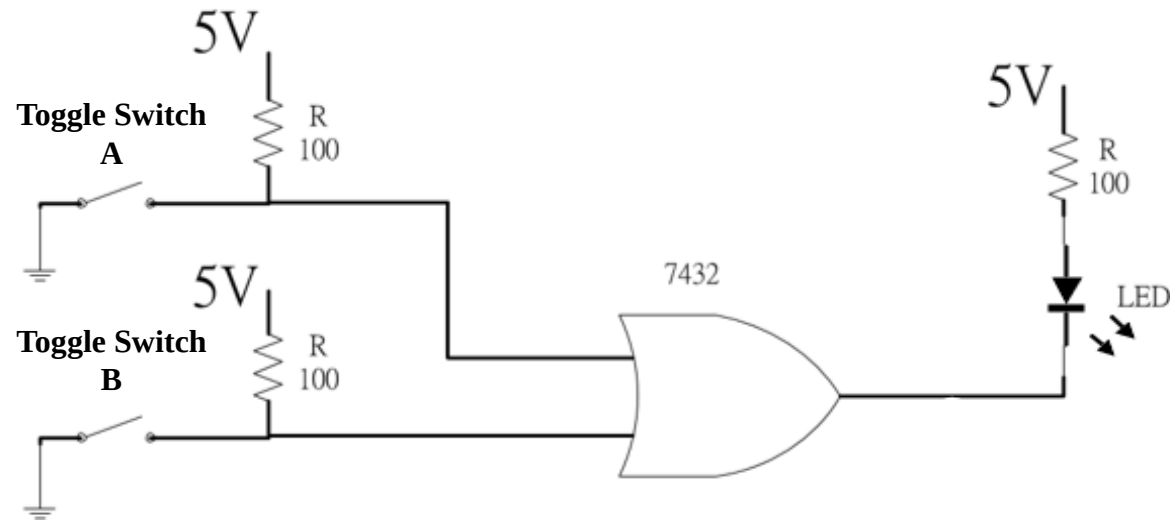
1. <https://jin-hua.com.tw/page/product/show.aspx?num=1366&lang=TW>

Circuit Application (Toggle Switch & LED)

- When the toggle switch is set to **ON**, the LED **lights up**.
- When the toggle switch is set to **OFF**, the LED **does not light up**.



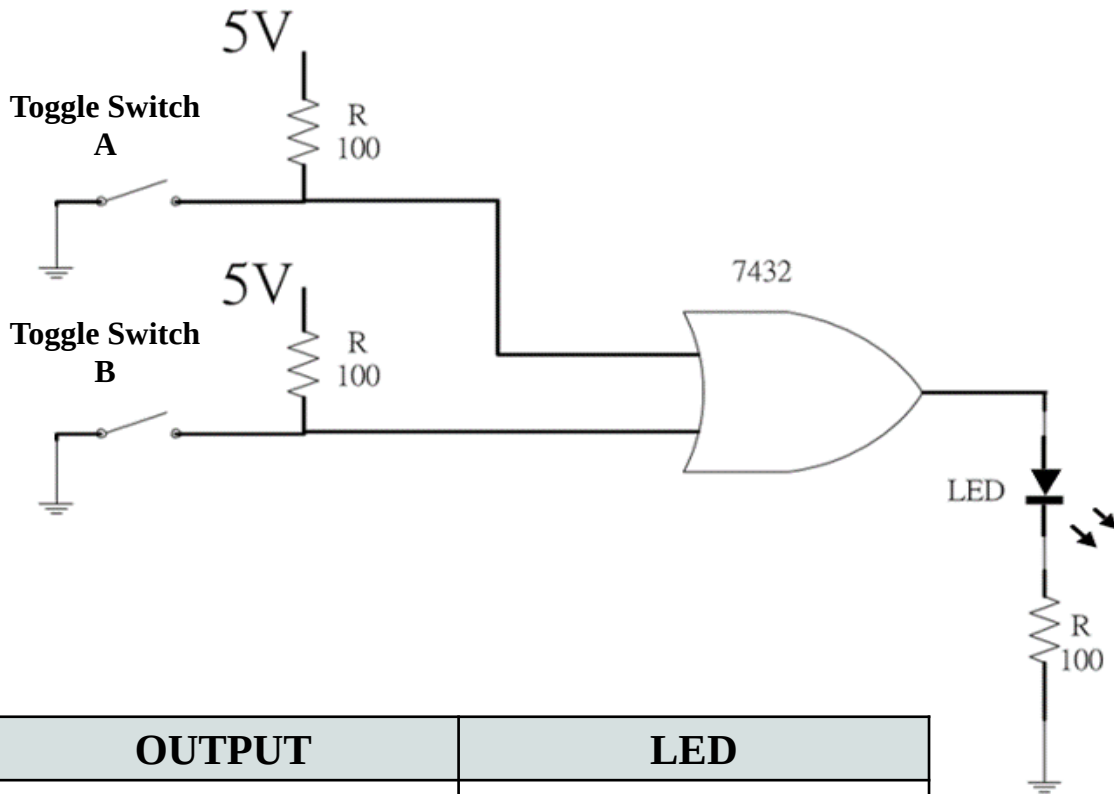
Practical Operation with 7432 (1/6)



OUTPUT	LED
0	on
1	off

Truth Table		
A	B	A+B
0	0	0
0	1	1
1	0	1
1	1	1

Practical Operation with 7432 (2/6)



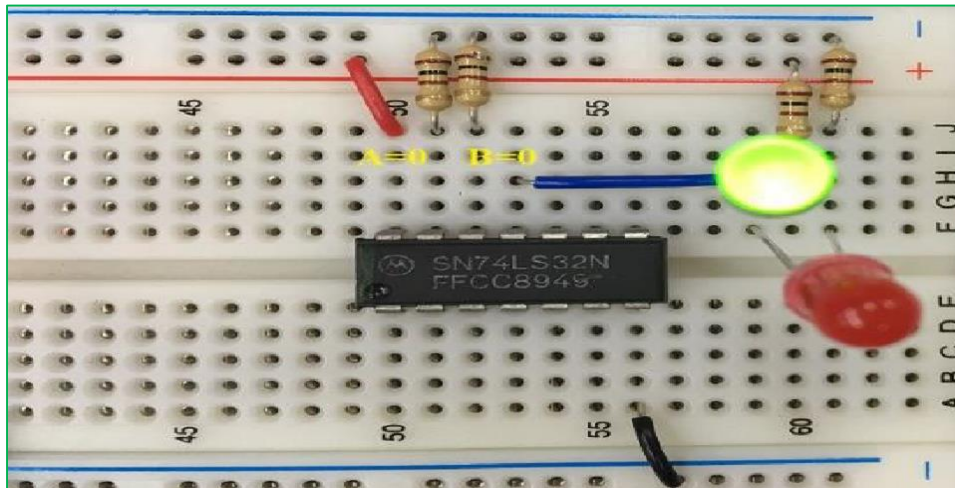
OUTPUT	LED
0	off
1	on

Truth Table		
A	B	A+B
0	0	0
0	1	1
1	0	1
1	1	1

Practical Operation with 7432 (3/6)

Truth Table (Measured Voltage) Output with LED Connected		
A	B	A+B
0.01V	0.01V	0.01V
0.01V	4.95V	3.1V
4.95V	0.01V	3.1V
4.95V	4.95V	3.1V

Truth Table (Measured Voltage) Output without LED Connected		
A	B	A+B
0.01V	0.01V	0.01V
0.01V	4.95V	4.28V
4.95V	0.01V	4.28V
4.95V	4.95V	4.28V

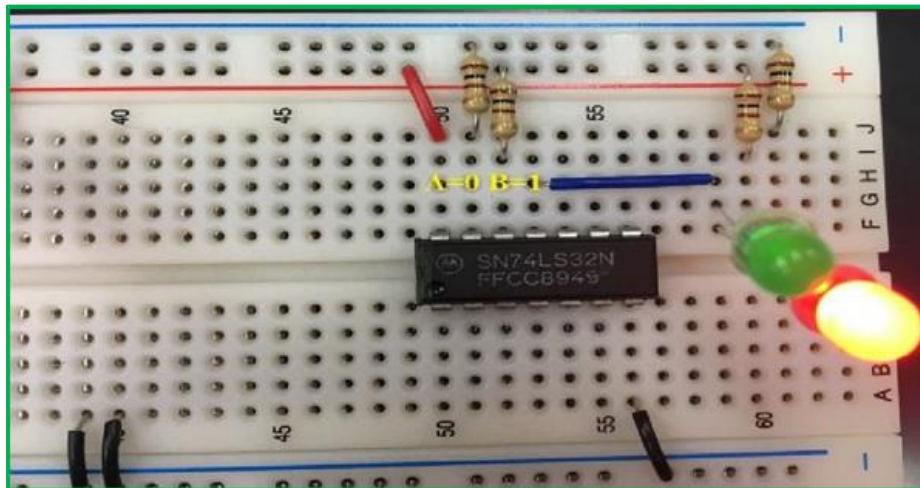


Truth Table (Logic)		
A	B	A+B
0	0	0
0	1	1
1	0	1
1	1	1

Practical Operation with 7432 (4/6)

Truth Table (Measured Voltage) Output with LED Connected		
A	B	A+B
0.01V	0.01V	0.01V
0.01V	4.95V	3.1V
4.95V	0.01V	3.1V
4.95V	4.95V	3.1V

Truth Table (Measured Voltage) Output without LED Connected		
A	B	A+B
0.01V	0.01V	0.01V
0.01V	4.95V	4.28V
4.95V	0.01V	4.28V
4.95V	4.95V	4.28V

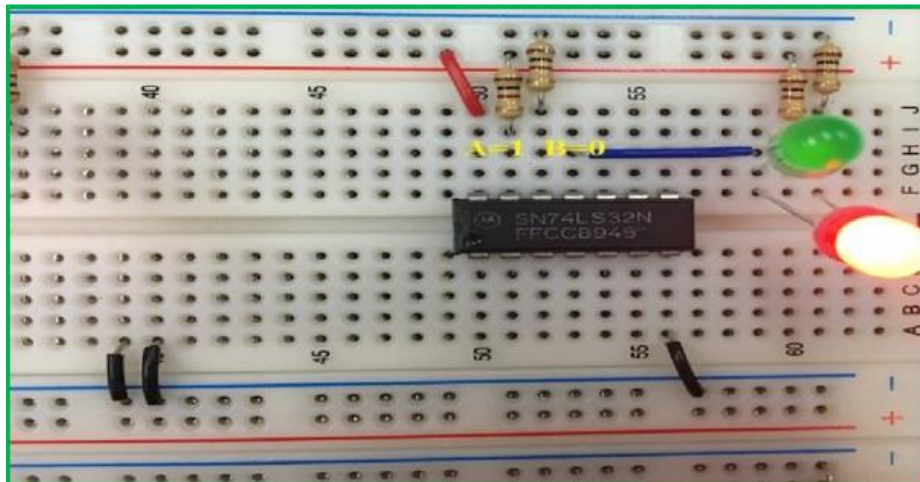


Truth Table (Logic)		
A	B	A+B
0	0	0
0	1	1
1	0	1
1	1	1

Practical Operation with 7432 (5/6)

Truth Table (Measured Voltage) Output with LED Connected		
A	B	A+B
0.01V	0.01V	0.01V
0.01V	4.95V	3.1V
4.95V	0.01V	3.1V
4.95V	4.95V	3.1V

Truth Table (Measured Voltage) Output without LED Connected		
A	B	A+B
0.01V	0.01V	0.01V
0.01V	4.95V	4.28V
4.95V	0.01V	4.28V
4.95V	4.95V	4.28V

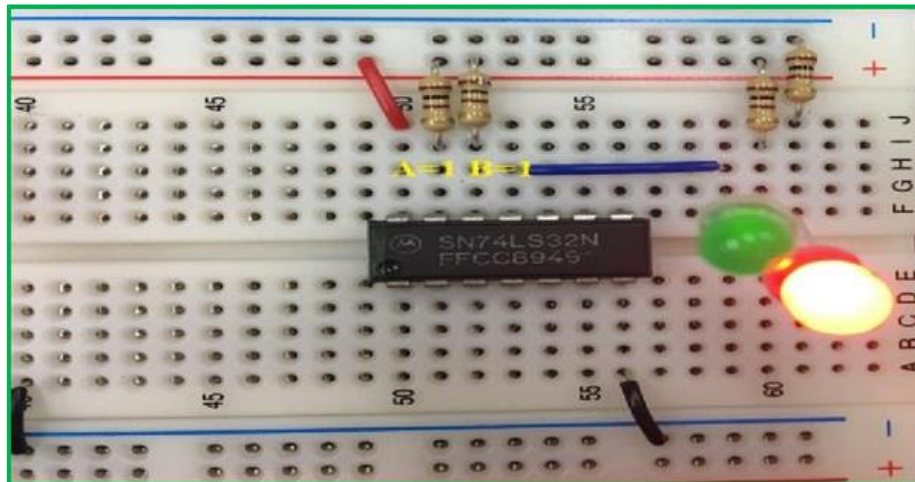


Truth Table (Logic)		
A	B	A+B
0	0	0
0	1	1
1	0	1
1	1	1

Practical Operation with 7432 (6/6)

Truth Table (Measured Voltage) Output with LED Connected		
A	B	A+B
0.01V	0.01V	0.01V
0.01V	4.95V	3.1V
4.95V	0.01V	3.1V
4.95V	4.95V	3.1V

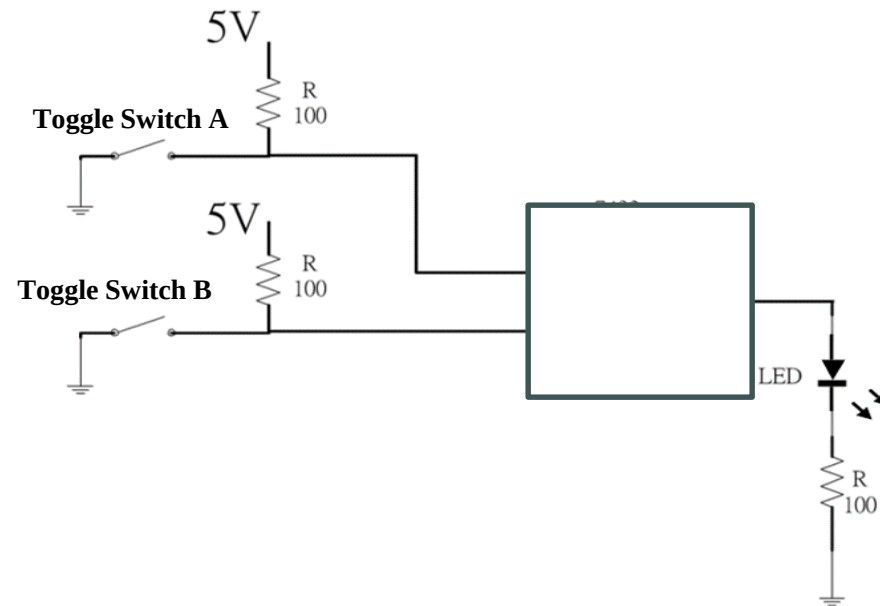
Truth Table (Measured Voltage) Output without LED Connected		
A	B	A+B
0.01V	0.01V	0.01V
0.01V	4.95V	4.28V
4.95V	0.01V	4.28V
4.95V	4.95V	4.28V



Truth Table (Logic)		
A	B	A+B
0	0	0
0	1	1
1	0	1
1	1	1

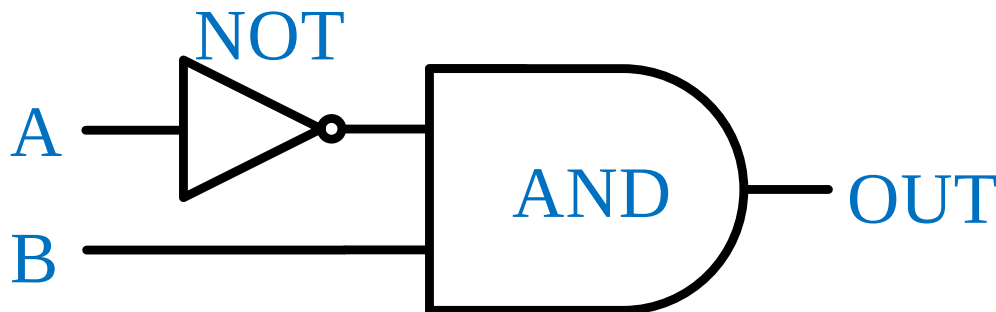
LAB 1-1

- Implement the circuits for **7404 (NOT)**, **7400 (NAND)**, and **7408 (AND)**.
- You need to provide the truth tables for all Measured Voltage with LED, Measured Voltage without LED, and Logic, respectively.
- For each case, photos of the results must be taken.



LAB 1-2

- Please implement the circuit shown below.
- You need to provide the truth tables for all **three** (Measured Voltage with LED, Measured Voltage without LED, and Logic, respectively).
- For each case, photos of the results must be taken.

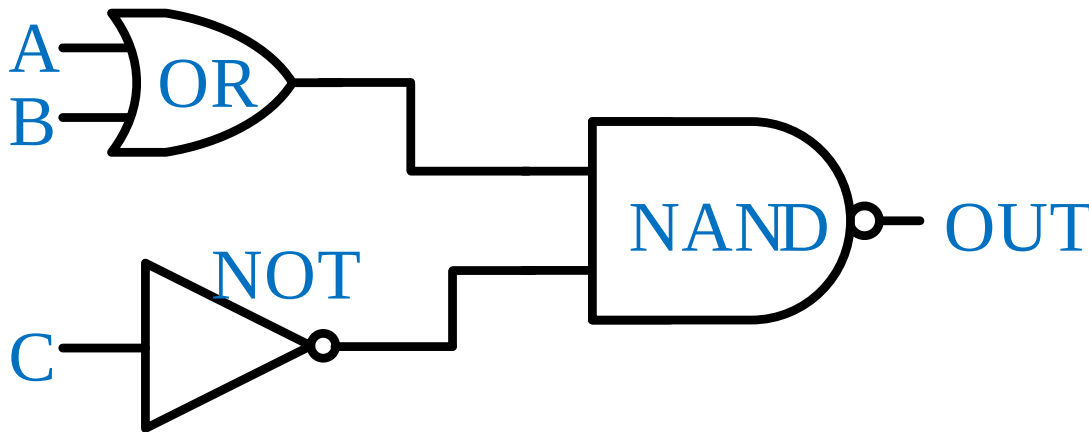


Truth Table (Measured Voltage)		
A	B	OUT

Truth Table (Logic)		
A	B	OUT

LAB 1-3

- Please implement the circuit shown below.
- You need to provide the truth tables for Measured Voltage with LED, Measured Voltage without LED, and Logic, respectively.
- For each case, photos of the results must be taken.



Truth Table (Measured Voltage)			
A	B	C	OUT

Truth Table (Logic)			
A	B	C	OUT